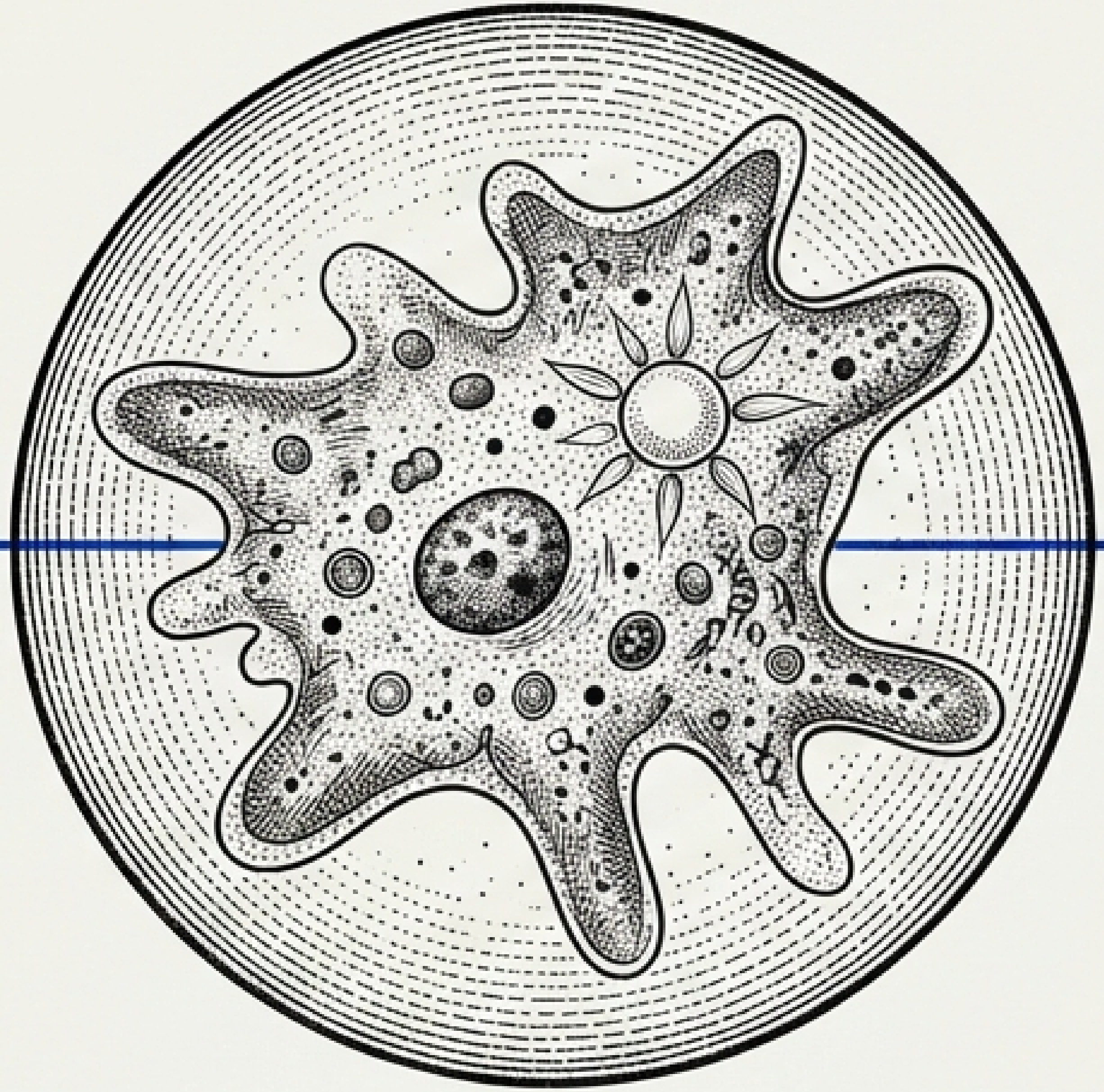
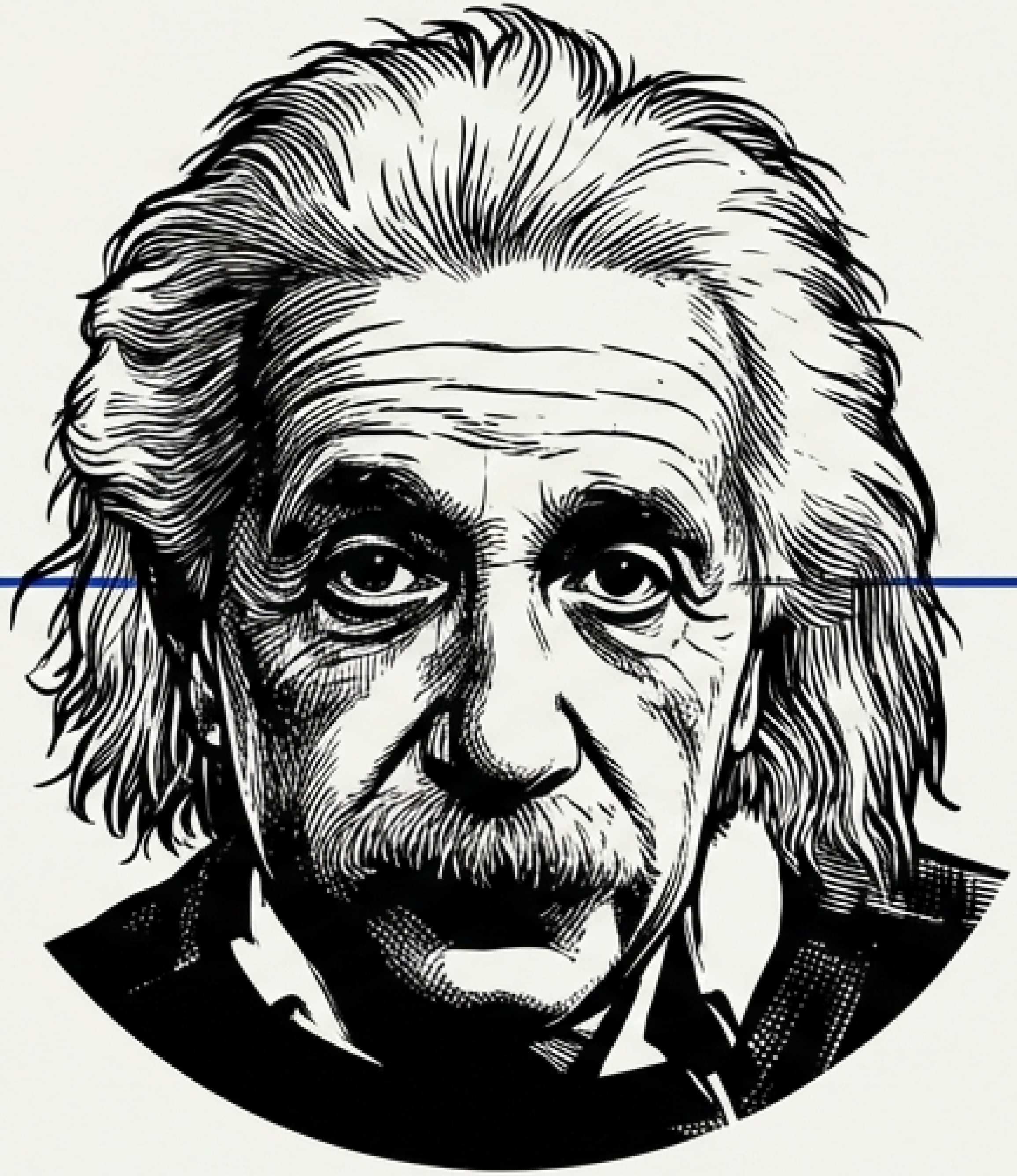
From Amoeba to Einstein

The Evolution of Objective Knowledge





Based on the work of Karl Popper | *Objective Knowledge: An Evolutionary Approach*

The Scandal of Philosophy

We all believe the sun will rise tomorrow. We bet our lives on regularities: that bread nourishes, that gravity holds, that day follows night.

The Conflict

- **Common Sense:** We know this because of repetition (induction).
- **Logic:** No number of white swans proves that all swans are white.

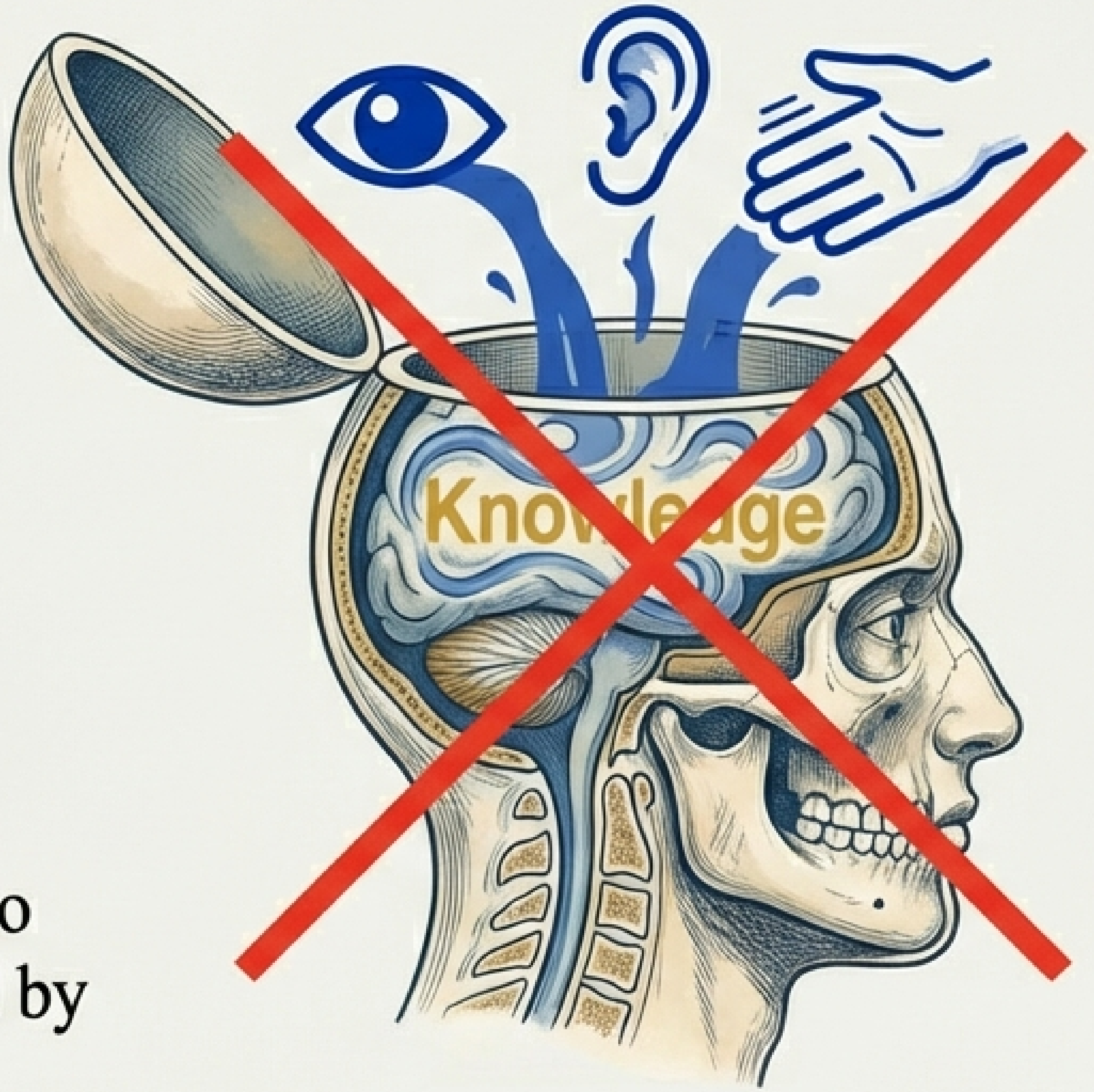
"The growth of unreason throughout the nineteenth century... is a natural sequel to Hume's destruction of empiricism." — Bertrand Russell

The Bucket Theory of the Mind

The Common Sense Error

The Traditional View (Aristotle to Locke):

1. Mind is an empty container (*tabula rasa*).
2. Knowledge enters through senses.
3. Accumulation of observations.
4. Distillation of theories.



The Verdict:

Popper argues this is biologically false. There is no 'pure' observation. Every observation is preceded by an interest, a problem, or a point of view.

Hume's Paradox

David Hume crushed the idea of Induction, creating a schizophrenia in Western thought.

The Logical Problem

Question: Are went of conrelling?
Are we justified in repeatent repeated instances to future conclusions?

Hume's Answer:
No. Repetition holds no logical power.

The Psychological Problem

Question:
Why do we expect the future to be like the past?

Hume's Answer:
Habit. We are conditioned by repetition, like animals.

The Consequence: Knowledge is reduced to irrational faith. 'There is no intellectual difference between sanity and insanity." — *Bertrand Russell*

Solving the Problem of Induction
"I have solved a major philosophical problem."

Popper accepts the Logical Result: We cannot justify the claim that a theory is true based on empirical reasons.

The Twist: He rejects the Psychological Result. We do not learn by passive repetition. We learn by active Conjecture and Refutation.

Old Model — Observation → Theory

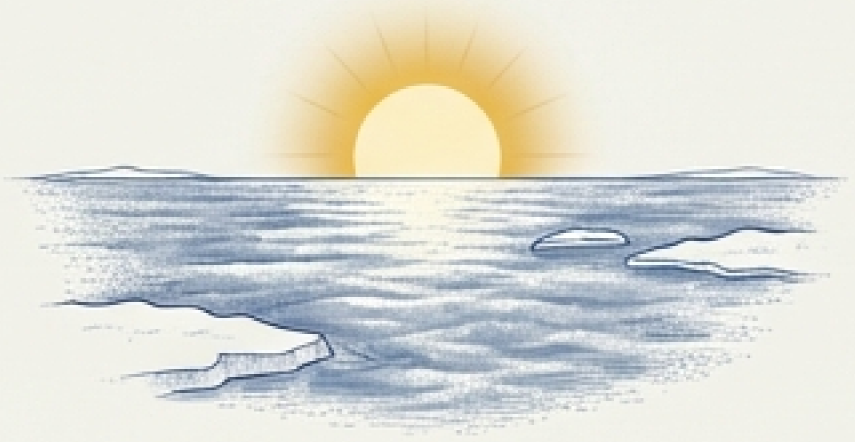
Popper's Model — Problem → Theory → Observation (Test)

'The regularities we try to impose are psychologically a priori.'

The Graveyard of Certainty

History proves that “Established Laws” are merely hypotheses waiting to be refuted.

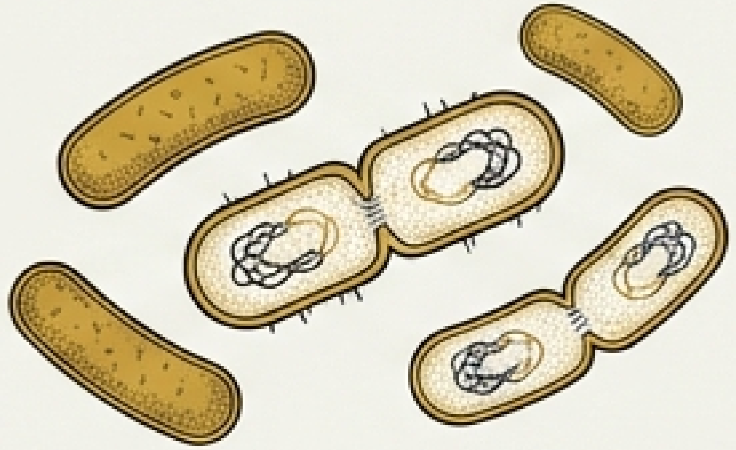
Exhibit A:
The Sun Rises



Belief: Sun sets every 24 hours.

Refutation: Pytheas discovers the “midnight sun”. The law was false.

Exhibit B:
All Men Are Mortal



Belief: Every creature dies.

Refutation: Bacteria multiply by fission; they do not die in the Aristotelian sense.

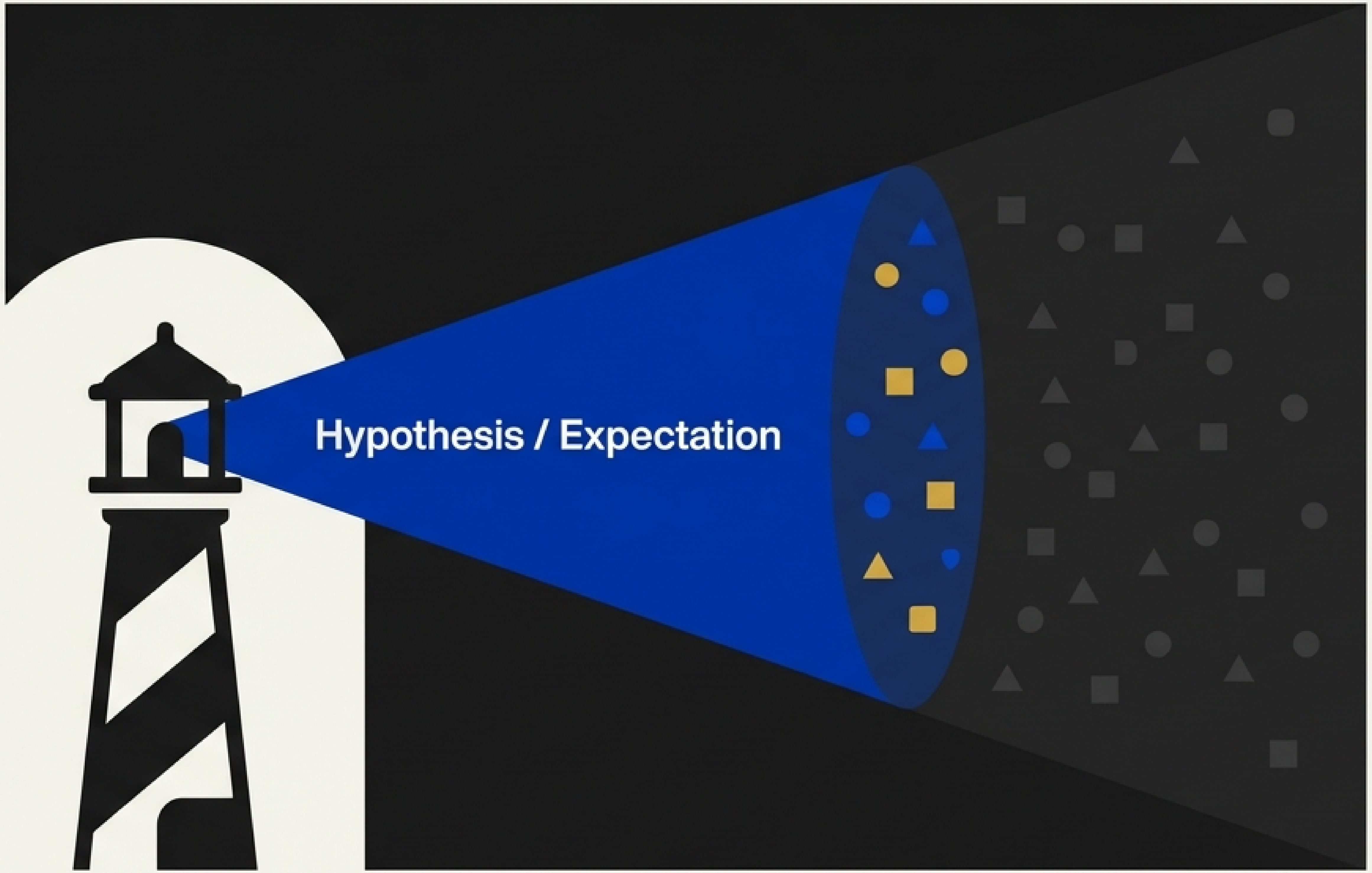
Exhibit C:
Bread Nourishes

Belief: Bread is safe.

Refutation: Ergotism outbreaks. Bread can be poison.

Takeaway: All laws are guesses. We can never prove them true, only false.

The Searchlight Theory



Knowledge does not flow in;
it is projected out.

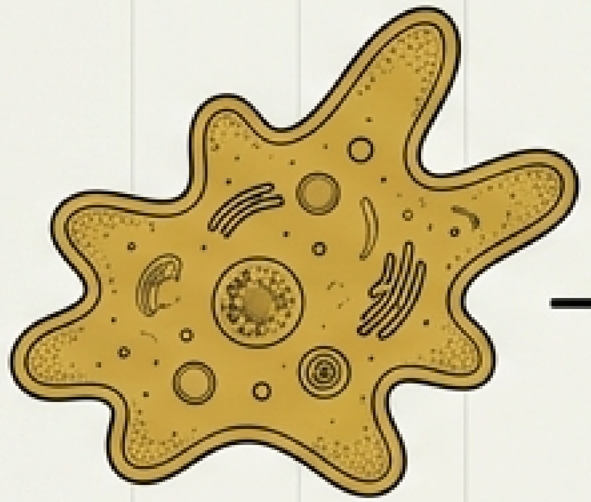
The Mechanism:

1. We start with an expectation (a hypothesis).
2. We use this hypothesis as a searchlight to scan reality.
3. We observe only what is relevant to our problem.

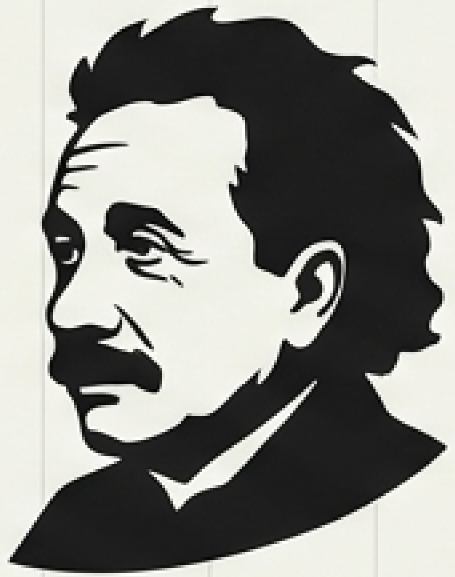
Key Insight: “Observations are secondary to hypotheses.”

From Amoeba to Einstein

Knowledge is a biological function. It is an organ of survival.



The Critical Difference: Einstein consciously seeks error elimination.



The Amoeba

Uses trial and error. Knowledge is genetic. If the theory fails, the amoeba dies.

Einstein

Uses trial and error. Theories are external (language). If the theory fails, the theory dies.

The Reality of the Sun

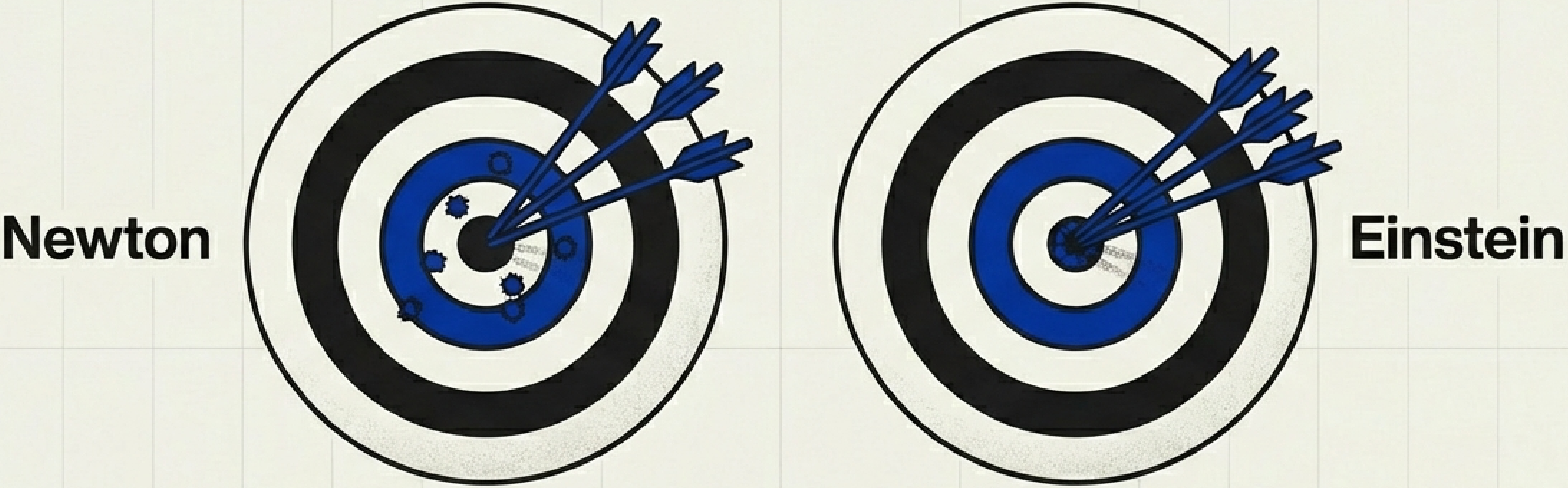
A Defense of Realism against Subjectivism

Winston Churchill's Rebuttal: Astronomers predict a black spot will pass the sun based on mathematics. We look, and the spot appears. We have independent testimony.

"The sun is real, and also it is hot—in fact as hot as Hell, and if the metaphysicians doubt it they should go there and see." — Winston Churchill

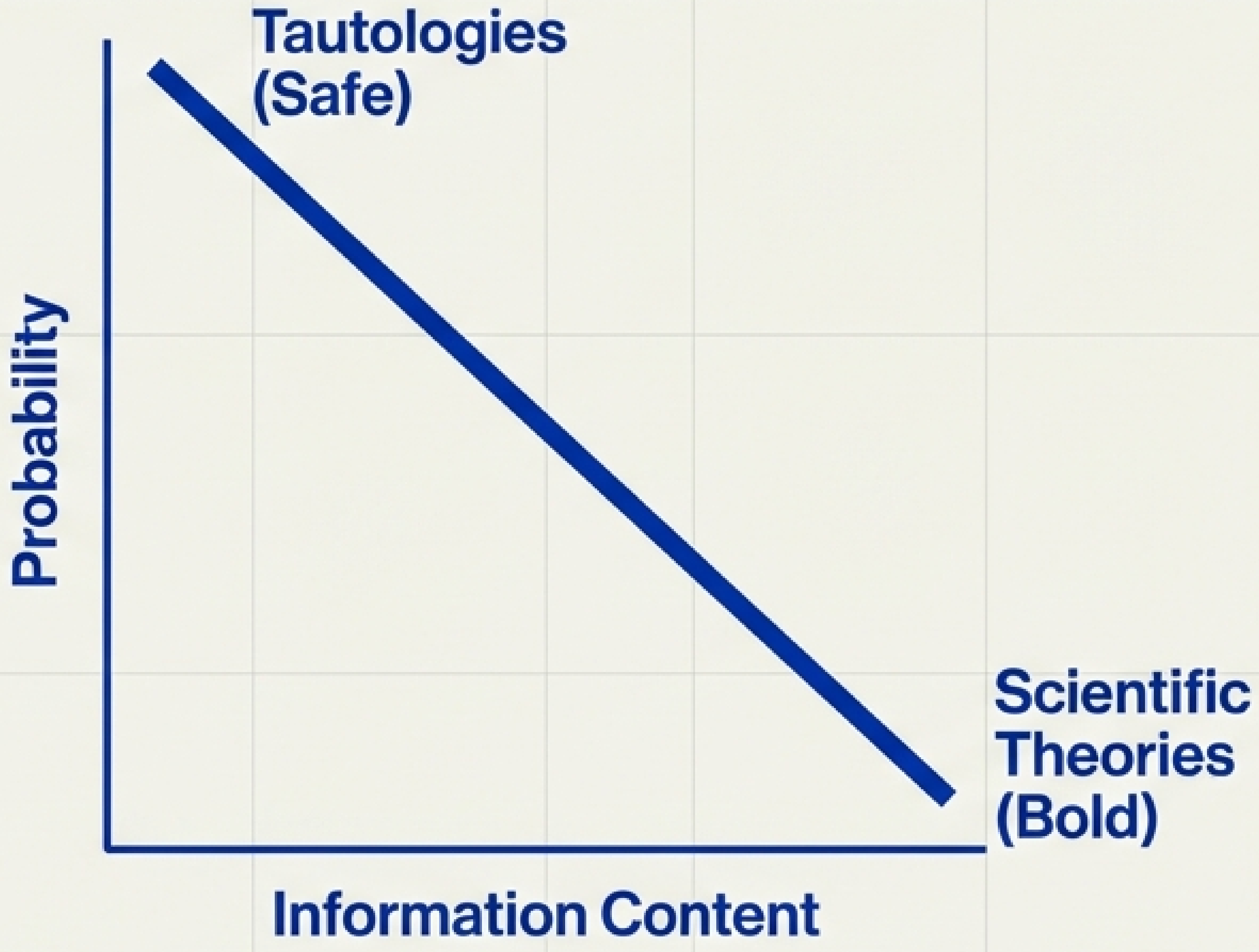
The Goal is Not Certainty

If all theories are guesses, why prefer one over another?
The Aim: Verisimilitude (Truth-likeness).



A theory is 'better' if it has higher truth content and lower falsity content than its competitor. We seek a better approximation.

The Merits of Improbability



We do not want highly probable statements.

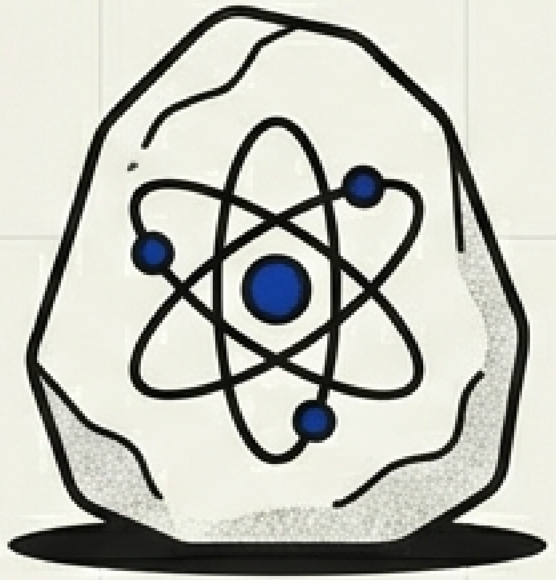
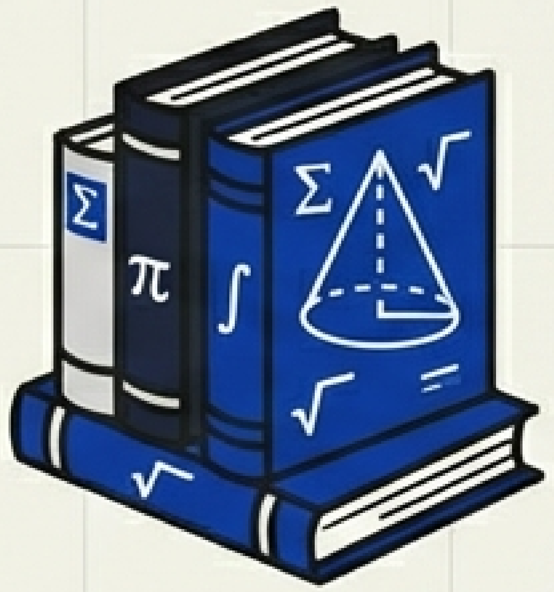
- "It will rain or not rain." (100% Probability, 0% Content)
- "Twice two is four." (True, but empty)

The Scientific Drive: We seek High Informative Content—which means Low Probability.

The Risk: The more a theory says, the more ways it can be wrong.

Science aims to be bold.

The Three Worlds

World 1	World 2	World 3
		
The Physical Universe. Things.	Subjective Experience. Mental states. Belief.	Objective Knowledge. Products of the human mind. Theories.

Crucial Insight: Once a theory is formulated (World 3), it becomes autonomous. It exists independently of the mind that created it.

The Autonomy of World 3

Prime Numbers

We discover problems in World 3 just as we discover mountains in World 1.

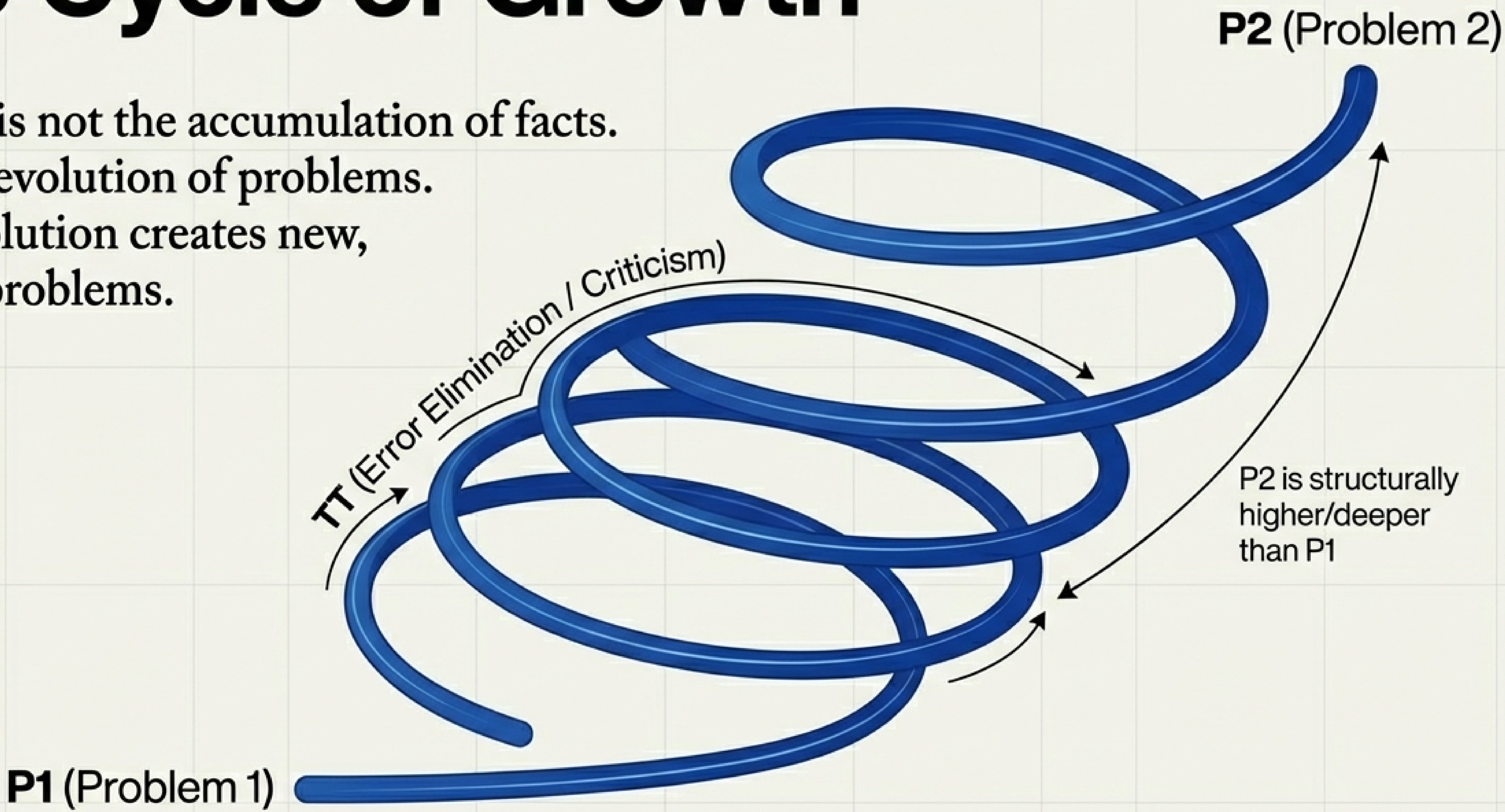
Example: The Prime Numbers.

- Humans invented arithmetic.
- But we discovered the infinite sequence of primes. We did not “invent” the problem; it emerged as an objective consequence of the system we built.

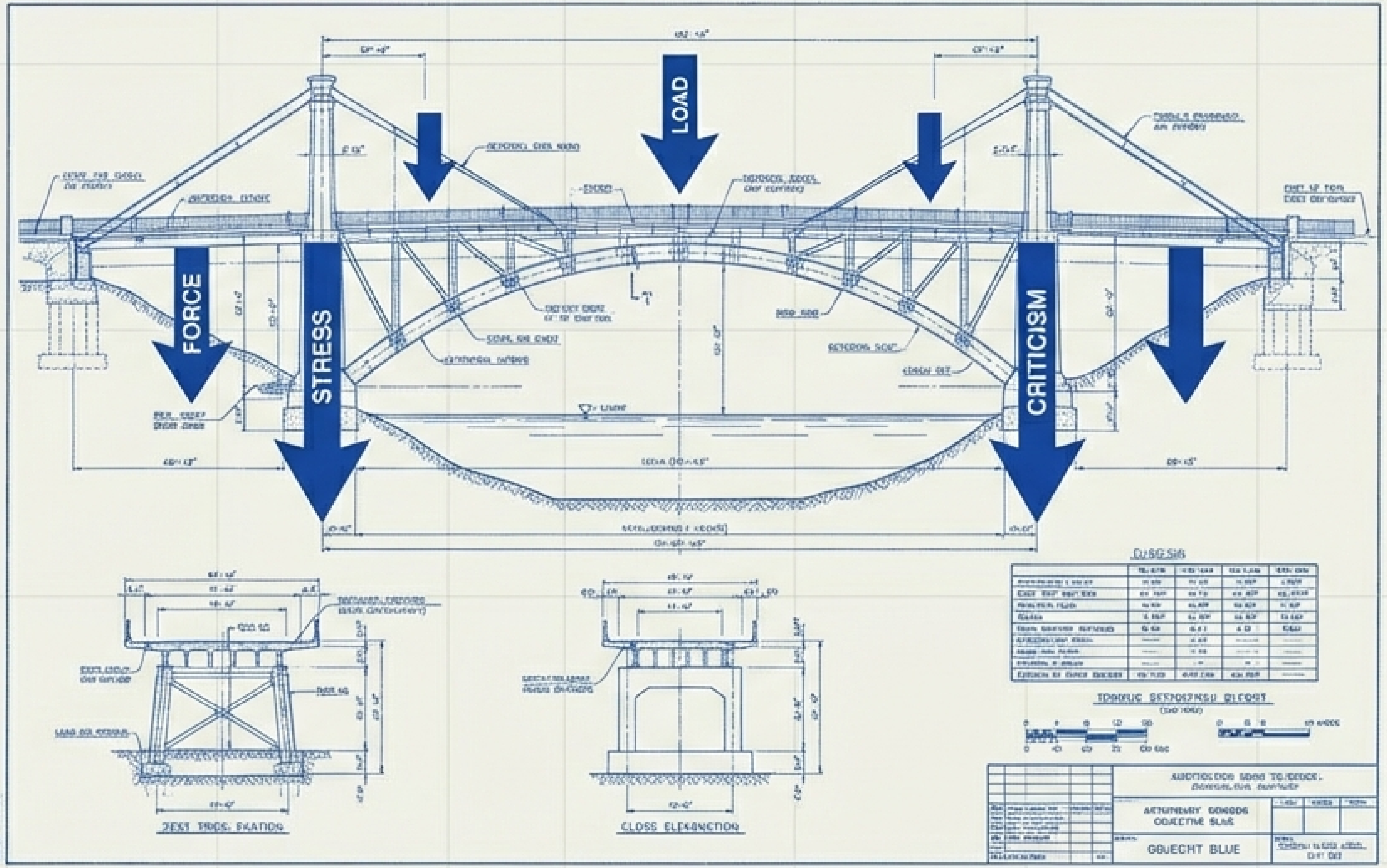
Knowledge can exist without a knowing subject. A library that no one reads still contains objective knowledge.

The Cycle of Growth

Science is not the accumulation of facts.
It is the evolution of problems.
Every solution creates new,
deeper problems.



Rationality in Action



If no theory is "true," how do we act?

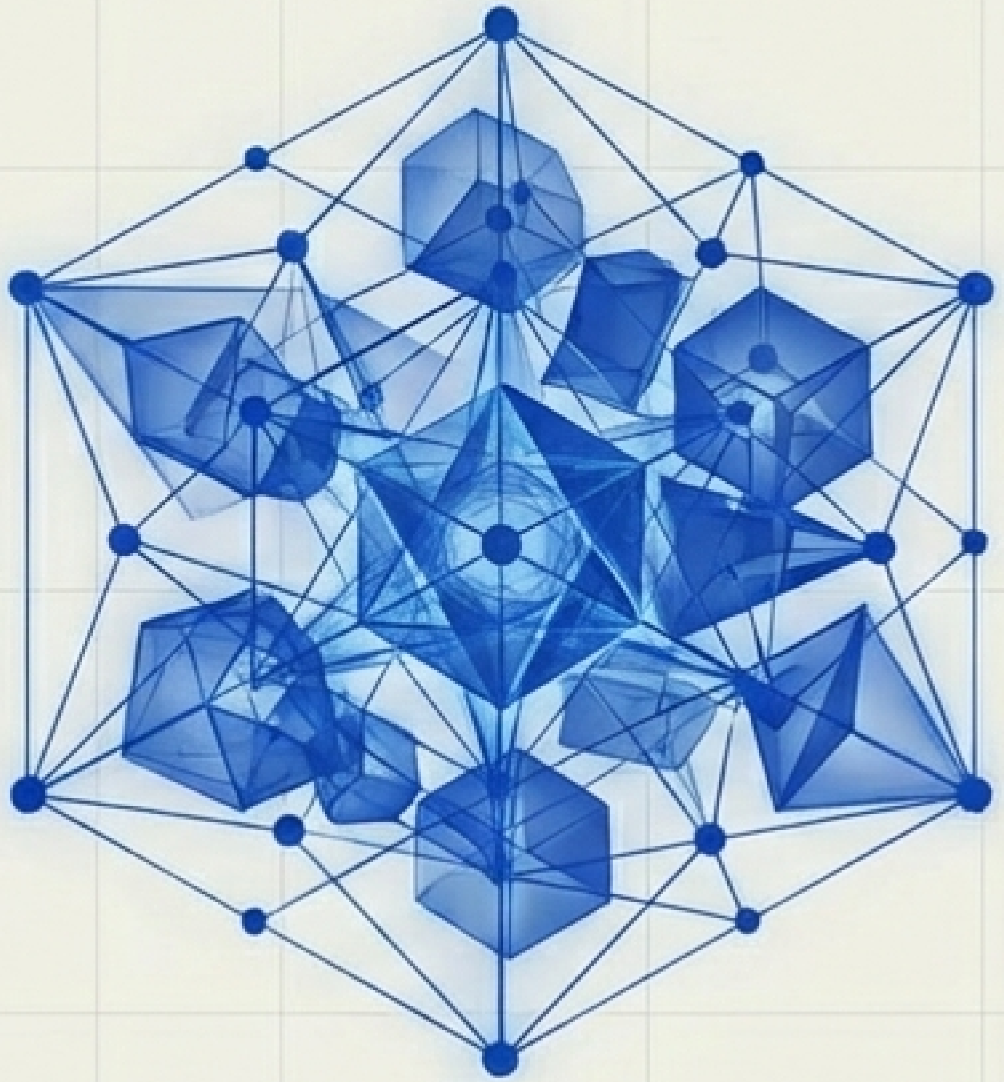
Pragmatic Preference:
We do not rely on any theory as certain (the bridge might fall). But we prefer the theory that is the Best Tested.

The Method:
1. Propose solutions.
2. Criticize them mercilessly.
3. Select the one that survives criticism.

Rationality is not justification.
Rationality is criticism.

The Duty of the Intellectual

“Aiming at simplicity and lucidity is a moral duty… lack of clarity is a sin, and pretentiousness is a crime.”



We are fallible.
We are searchers, not possessors of truth. But through the cooperation of bold conjecture and severe criticism, we can break out of the subjective bucket and build an **objective world of knowledge.**